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MILITARY SPECIFICATION

AIRPLANE STRENGTH AND RIGIDITY FLIGHT AND GROUND OPERATIONS TESTS

1. SCOPE

1.1 This specification covers the flight test requirements for the structural integrity flight and ground evaluation and demonstration of piloted airplanes.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-8860	Airplane Strength and Rigidity - General Specification for
MIL-A-8861	Airplane Strength and Rigidity, Flight Loads
MIL-A-8862	Airplane Strength and Rigidity, Landplane Landing and Ground Handling Loads
MIL-A-8865	Airplane Strength and Rigidity - Miscellaneous Loads
MIL-A-8868	Airplane Strength and Rigidity, Data and Reports

PUBLICATION

WADC Technical Report 57-162	Preparation of Strength Summary and Operating Restrictions Reports (AD 118326)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

FSC 1510

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National Advisory Committee for Aeronautics

NACA Report 1178 Calibration of Strain-Gage Installations in Aircraft Structures for the Measurement of Flight Loads

(Application for copies should be addressed to the Defense Document Center, Defense Supply Agency, Cameron Station, Alexandria, Virginia 22314.)

3. REQUIREMENTS

3.1 Types of programs. The structural flight test programs specified herein are:

a. Flight and ground operations load survey - To measure inflight and ground operational loads for substantiating the structural design loads analysis and to demonstrate the structural integrity of the airplane for the critical loading conditions.

b. Flight and ground operations demonstration - To demonstrate the structural integrity of the airplane for the critical loading conditions.

c. Dynemic response tests - To measure the elastic response characteristics of the airplane structure to dynamic load inputs for substantiating the dynamic loads analysis.

The procuring activity will designate which program or programs will be applied to each specific airplane weapon system or model thereof.

3.2 Airplanes for test

3.2.1 The airplane(s) to be tested shall be designated by the procuring activity. Unless otherwise specified, the second airplane produced of each airplane model procured shall be instrumented and flight tested by the airplane contractor personnel in accordance with the requirements and intent of this specification. An additional airplane, sufficiently late in the production program to insure obtaining the final configuration, shall also be designated for structural flight test and shall be instrumented as specified by the procuring activity. This airplane shall serve as a standby for the initial test airplane and as the test article for the final tests in the event that it becomes impractical (because of modifications or other reasons) to use the initial airplane for final tests.

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3.2.2 The structural flight test airplane shall be structurally, inertially, and aerodynamically similar to the design presented in the structural analysis and drawings. Configuration, materials, and quality of workmanship shall be the same as for the service aircraft. Significant modifications arising during the development program of the airplane shall be incorporated on the test airplane, or a later airplane shall be flight tested to cover the modifications.

3.3 Official witness. The tests required by this specification shall be witnessed at the discretion of the procuring activity.

3.4 Instrumentation requirements. The following general requirements shall be used to determine the minimum acceptable level of instrumentation for a structural flight and ground operations test. The contractor shall recommend the specific instrumentation and location of same, with justification for approval by the procuring activity, to meet the requirements applicable to a particular aircraft design. Sufficient instrumentation shall be provided such that the measurements obtained will allow valid comparisons of the test data with the appropriate loads analysis and design conditions.

3.4.1 Load instrumentation. Structural loads may be measured by the strain gage method, the pressure survey method, or both. Accelerometers, thermal instrumentation, and external store instrumentation shall also be used. The loads instrumentation shall be capable of measuring the following:

a. Flight loads - The spanwise distribution of bending moment, shear, and torsion on the primary load-carrying surfaces of the aircraft shall be measured. The other parts of the airplane to be considered for loads measurement shall include, but not be limited to; auxiliary lift and drag devices, landing gear doors, weapon bay doors, control surfaces and devices, hatches when open, critical drive rods (i.e., inlet), tail bumper, and tail hook. When gust loads are measured, a gust-sensing device shall be installed for the measurement of gust velocities.

b. Ground loads - The vertical, drag, side, and torsion loads on the landing gear shall be measured. Loads on other parts of the aircraft (which experience significant loads during ground operation) shall be measured utilizing the flight loads instrumentation to the maximum degree possible and practical.

3.4.1.1 Strain gage method. Wing load distribution measurements by the strain gage method (one wing) shall include at least four stations for spanwise bending moment, shear, and torsion to cover the wing root, two midspan, and wingtip region(s) on small-to-medium-size airplanes (semispan

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along quarter-chord line, 50 feet or less); and at least six stations for spanwise bending moment, shear, and torsion measurements to cover the wing root, four midspan, and wingtip region(s) on medium-to-large-size airplanes (semispan along quarter-chord line greater than 50 feet). Unless otherwise agreed, spanwise distribution shall be obtained on the right wing. In such instances, however, instrumentation shall be provided for obtaining at least the total loads (bending moment, shear, and torsion) on the opposite wing to facilitate the determination of unsymmetrical loads. Tail loads measurements by the strain gage method shall include at least the total loads (bending moment, shear, and torsion) at the root of each horizontal and vertical surface. Vertical and side bending moments and shears shall be measured for at least one forward fuselage station and one aft fuselage station. Since landing gear configurations may vary greatly between aircraft, specific strain gage requirements cannot be specified. However, the landing gear and backup structure shall be sufficiently instrumented with strain gages to measure vertical, drag, side, and torsion loads. For certain aircraft, additional strain gages may be required as agreed upon between the contractor and procuring activity.

3.4.1.2 Pressure survey method. Unless otherwise agreed, wing load distribution measurements by the pressure survey method (one wing) shall include at least five stations on small-to-medium-size airplanes and at least seven stations on medium-to-large-size airplanes to allow determination of bending moment, shear, and torsion measurements to cover the wing root midspan, and wingtip regions. Tail load measurements by the pressure survey method shall include measurements at a minimum of three stations on each horizontal and vertical surface on small-to-medium-size airplanes and measurements at a minimum of five stations per surface on medium-to-large-size airplanes to allow determination of bending moment, shear, and torsion at the root of each horizontal and vertical surface. Sufficient fuselage pressure measurements shall be made to allow determination of the vertical and side bending moments and shears for at least one forward fuselage station and one aft fuselage station.

3.4.1.3 Accelerometers. The load instrumentation shall include sufficient accelerometer coverage (in addition to the requirements specified in 3.4.2) over the airplane to permit determination of the inertia load distributions as required to correlate with or substantiate the static and dynamic loads analysis. Accelerometers shall be placed at the following locations as applicable or as specified by the procuring activity.

a. Vertical accelerations on both the right and left wing and horizontal tail tips and wing midspan regions

b. Lateral accelerations at the vertical tail tip(s)

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c. Both vertical and lateral acceleration at the nose and tail of the airplane

d. Three axes rectilinear accelerations and three axes angular rates near the center-of-gravity of large concentrated mass items (engines, stores, et cetera) not buried in the fuselage or wing root

e. Accelerations in the direction of primary structural response on auxiliary lift and drag devices

f. Landing gear doors, weapon bay doors, lift engine and fan doors

g. Control surfaces and devices

h. Hatches, when open

i. Tail bumper

j. Tail hook.

3.4.1.4 Thermal instrumentation. The instrumentation for thermal loads shall consist of systems capable of measuring skin temperature distributions and internal structural temperatures when specified by the procuring activity. Sufficient instrumentation and measurements shall be provided to allow valid comparisons to be made with the appropriate design loads analyses.

3.4.1.5 External store instrumentation. Instrumentation shall be installed for determination of the loads at the interface of the store pylon with the airframe or the store with the airframe to substantiate the structural flight integrity of the airplane/store combination. Loads measurements shall include but not be limited to vertical, side, and drag load and pitching, yawing, and rolling moment as applicable.

3.4.2 Instrumentation for test condition parameters. The measurements obtained from this instrumentation shall be used to prove compliance with test requirements and to evaluate transient response characteristics of the airplane. Instrumentation to measure the following parameters shall be utilized, as applicable. Items provided under the loads instrumentation requirements specified in 3.4.1 need not be duplicated. (Note: This assumes that suitable instrumentation is provided for the pilot.)

a. Airspeed and Mach number

b. Pressure altitude

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c. Normal, lateral, and longitudinal linear acceleration at the center-of-gravity of the airplane

d. Aileron, elevator (or stabilator), and rudder control forces (corresponding forces on unconventional airplanes)

e. Aileron, elevator (or stabilator), and rudder positions (corresponding surfaces on unconventional airplanes), and dive brake position, if applicable

f. Pilot's control positions corresponding to the measurements specified in 3.4.2e

g. Aileron, elevator (or stabilator), and rudder hinge moments (corresponding surface hinge moments on unconventional airplanes)

h. Roll, pitch, and yaw angular rates and accelerations at the center-of-gravity

i. Angle of attack, sideslip, and bank

j. Airplane weight, moments of inertia, cross products of inertia, and center-of-gravity (computed)

k. Stress and temperature measurements on critical structural components when specified by the procuring activity. Provision shall be made to provide necessary reference strain or zero reading.

l. Stagnation and ambient air temperature

m. Twist and bending deflections (by camera and targets), when specified by the procuring activity

n. Airplane configuration (auxiliary lift and drag device positions, landing gear positions, airplane geometry, stores, engine positions, et cetera, as applicable)

o. Airplane sink speed

p. Landing and taxi ground speed

q. Landing gear strut positions

r. Oleo stroke and air pressure

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s. Tire inflation pressure

t. Other measurements as specified by the procuring activity.

3.4.3 Responsibility for instrumentation. Unless otherwise specified, the contractor shall be responsible for the installation and calibration of the instrumentation required to conduct the approved structural flight test program. For the strain gage method, the calibration procedures presented in NACA Report 1178 are acceptable. The contractor shall also be responsible for the recording, data reduction, and analysis of the required measurements. The instrumentation, installation, calibration, data-handling equipment, and the methods of data reduction and analysis shall be specified and justified by the contractor and approved by the procuring activity.

3.5 Failure to attain test maneuver requirements. When stability and control or other aerodynamic difficulties prevent the attainment of any test maneuver requirement, the procuring activity will decide whether the airplane shall be aerodynamically modified and the tests continued to completion or a deviation granted from the unattained requirement.

3.6 Reinforcements. The structural flight tests shall be discontinued for repair or reinforcement of the airplane whenever such action is mutually agreed to by the procuring activity and the contractor. Any structure incurring failure shall be adequately repaired. Any part showing objectionable permanent set during the structural flight test shall be reinforced to withstand the required maneuvers or replaced with a newly designed part. The requirement for repeating tests with reinforced structures shall be determined by the procuring activity.

3.7 Operating limitations. Structural operating limitations and substantiating data shall be provided by the contractor for each type, series, model, and configuration of airplane in the form of strength summary and operating restriction reports in accordance with MIL-A-8868, WADC Technical Report 57-162, or the appropriate contractual document. In the interest of flight safety, pending substantiation of structural integrity by static tests and flight tests, airplanes are normally rated as having that structural capability necessary to sustain 80 percent of design limit load on primary structural components if the structural design loads and stress analysis have been submitted to and approved by the procuring activity. This 80-percent limitation shall be applied to all test and service aircraft of the same model. Permission to exceed this limitation for test aircraft shall be approved by the procuring activity.

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3.7.1 Eighty percent operating restrictions. In applying the 80-percent limitation to operational services, airplane symmetrical maneuver load factors of +2.0g and 0.0g shall be provided as the minimum acceptable values for fleet use.

3.7.2 Release of structural limitations. The release of structural operating limitations for test and service inventory aircraft to operate at the 100 percent limit load level shall be contingent upon successful completion of the flight loads survey program and the submittal to and approval by the procuring activity of all test results and analyses pertaining thereto (loads and stress analyses, static tests, flight tests, and strength summary and operating restrictions report) and structural modification resulting therefrom. The structural flight test airplane only shall proceed with testing at the 100 percent limit load level upon completion of the 80 percent structural flight test program and ultimate load static tests subject to the approval of the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Test program requirements. The structural flight test condition requirements for the following types of test programs are specified herein.

a. Flight and ground operations loads survey. (4.2)

b. Flight and ground operations demonstration. (4.3)

c. Dynamic response tests. (4.4)

4.1.1 The requirements specified in 4.2 through 4.4.4 cover all types of airplane weapon systems. In preparing a test program for a particular airplane, the contractor shall analyse all requirements which are applicable to that particular airplane. These recommendations shall be submitted to the procuring activity for approval.

4.2 Flight and ground operations load survey. The test conditions for this survey shall cover the various operating regimes (gross weight, load factor, speed, altitude, etc) of the particular airplane such that the measured loads data may be used to correlate with and substantiate the airplane design loads analysis. The contractor shall select from the following paragraphs those test conditions, test loadings, configurations, types of maneuvers, and load factors, etc, required for the particular airplane so that they will agree with, and will be based upon, the structural design parameters as determined in accordance with MIL-A-8861, MIL-A-8862, MIL-A-8865, the approved design criteria report, or the contract end item

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specification, whichever is contractual. Test speeds and altitudes shall be chosen to cover all conditions within the design speed-altitude envelope. This survey shall be conducted in the following three phases:

a. Initial phase flight tests to 80 percent limitations (4.2.1)

b. Final phase flight tests to 100 percent limitations (4.2.2)

c. Ground operating loads testing (4.2.3).

4.2.1 Initial phase flight tests. The following test maneuvers and condition parameters shall be included as applicable by the contractor in preparing the test program. Eighty percent of design limit load on any primary structural component of the airplane shall not be exceeded during this phase of the test program.

4.2.1.1 Flight load survey maneuvers at high, intermediate, and low altitudes. The flight load survey maneuvers at high, intermediate, and low altitudes shall consist of the following:

a. Normal symmetrical pullup (4.5.1).

b. Normal symmetrical pushdown (4.5.2).

c. Normal uncoordinated rolling pullout (4.5.3).

d. Normal uncoordinated 360-degree roll (4.5.4).

e. Steady-state yaw maneuver (4.5.5).

4.2.1.1.1 Altitudes. The maneuvers specified in 4.2.1.1 shall be performed at three altitudes, as applicable. The altitudes shall be low, intermediate, and high consistent with the airplane flight envelope and considering safety of flight of the airplane and crew. The minimum altitude need not be lower than 5,000 feet above terrain.

4.2.1.1.2 Test speeds. Five test speeds (see 6.2.1) at each altitude shall be included as applicable, for all maneuvers specified in 4.2.1.1, except the pushdown maneuver which requires a minimum of three test speeds. The lowest test speed for the normal symmetrical pullup, the normal symmetrical pushdown, and normal uncoordinated rolling pullout maneuvers shall be the minimum design speed for the limit load factor for the particular maneuver at the required test altitude and required weight condition. The lowest test speed for the normal uncoordinated 360-degree roll maneuver shall be the same as that for the normal uncoordinated

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rolling pullout maneuver. The lowest test speed for the steady-state yaw maneuver shall be the limit flaps-down speed. The highest test speed is the maximum design speed. When a critical design analytical condition occurs at an intermediate speed for the particular test altitude, the test speeds shall be arranged to include such a condition.

4.2.1.1.3 Loadings and configurations. The normal symmetrical pullup, the normal symmetrical pushdown, and the normal uncoordinated rolling pullout maneuvers shall be conducted with the airplane internal loading considered critical for the wing in the clean configuration, unless fixed external stores of pylons are a part of the design, and with such external store configurations and loadings considered to significantly affect the wing or horizontal tail loads. The normal uncoordinated 360-degree roll maneuver shall be conducted with the loadings and configurations considered critical for vertical tail loads. The steady-state yaw maneuver shall be conducted with the airplane internal loading considered critical for vertical tail loads in the clean configuration and with such external store configurations and loadings considered to significantly affect the sideslip angles. Where extended dive brakes, dive flaps, etc, are expected to produce significant changes in the wing or horizontal tail loads, the applicable maneuvers shall be repeated with these control devices extended. The effects of flaps or other high-lift devices shall also be determined by performing the applicable maneuvers within the design speed-altitude envelope of these devices. The altitude (4.2.1.1.1) and speed (4.2.1.1.2) requirements may be modified for the flap configuration survey at the discretion of the procuring activity.

4.2.1.2 Design criteria flight conditions. These tests shall include all maneuvers required for the particular type and class of airplane at the speed-altitude combinations indicated to be critical by analytical calculations. Critical design analytical conditions previously performed during flight load survey maneuvers (4.2.1.1) shall not be repeated. The maneuvers specified under 4.5 shall be performed as applicable.

4.2.1.2.1 Altitudes. All maneuvers shall be conducted at the altitudes indicated to be critical by analytical calculations. However, where these altitudes occur at sea level or near sea level, the test altitude shall be selected as being the lowest altitude possible considering safety of flight of the airplane and crew. The minimum altitude need not be lower than 5,000 feet above terrain.

4.2.1.2.2 Speeds. All maneuvers shall be so conducted as to obtain the required test conditions at the speeds indicated to be critical by analytical calculations.

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4.2.1.2.3 Loadings and configurations. The maneuvers specified in 4.2.1.2 shall be performed at the airplane gross weight and internal loading indicated to be critical by analysis. Where external store configurations and loadings significantly affect airplane loads, the applicable maneuvers shall be repeated with the most critical store configuration. In any instance where extended dive brake, dive flaps, spoilers, in-flight operable thrust reversers, etc, are expected to produce significant airplane load changes, the applicable maneuvers shall be repeated with these devices extended.

4.2.1.2.4 Maneuvers for critical external stores support. Test conditions for the configurations and maneuvers determined by analysis to produce maximum loads on external store supports shall be performed.

4.2.1.2.5 Maneuvers for critical aerodynamic heating. Maneuvers for critical aerodynamic heating shall be determined by analysis.

4.2.1.2.6 Miscellaneous loading conditions. Response to store thruster firings (ejections), braking wheels in air, extension and retraction of landing gear, buffet, and other miscellaneous load conditions shall be measured. These data shall be analyzed to substantiate that these various loading conditions will not result in the design loads being exceeded.

4.2.1.2.7 Test required to define suspected new critical conditions. If analysis of the flight loads data indicates the possibility of new critical conditions not previously investigated, additional tests shall be conducted.

4.2.2 Final phase flight tests. The final phase flight tests shall be performed after completion of the initial phase flight tests, the ultimate load static tests, and with the specific approval of the procuring activity. The following test conditions shall be included, as applicable, by the contractor in preparing the test program. Buildup maneuvers shall be flown as necessary to insure not exceeding the 100 percent design limit load of the aircraft.

a. Conditions indicated to be critical by analytical calculations, static test data, and initial phase flight test data.

b. Any initial phase tests deferred because of speed restrictions, nonavailability of production engines, external stores, critical loadings, etc. These tests shall be performed in a buildup manner to the 80 percent limitations specified in 4.2.1 before performing the 100 percent limit load tests.

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c. Conditions specified by the procuring activity to gain additional substantiation for the aircraft structural limitations.

4.2.3 Ground operations load survey conditions. These tests shall include all ground operating conditions required for the particular type and class of airplane which are indicated to be critical or near-critical by analytical calculations. These tests shall be conducted after the successful completion of the ultimate static test for all ground loading conditions. The test program shall include the following test conditions, as applicable:

4.2.3.1 Discrete bump or dip tests. Test conditions shall encompass a sufficient number of aircraft weights, velocities, center-of-gravity locations, bump or dip amplitudes and wave lengths, et cetera, to substantiate the loads analysis and demonstrate the critical loading conditions. The critical wavelengths shall be determined by considering the taxi, takeoff, and landing speeds of the airplane and by the frequencies of the rigid body and flexible modes of the airplane for each weight and/or store configuration. Tests shall be conducted in a systematic manner to build up to the critical conditions with the final test conditions determined by analysis and preliminary testing. The bumps or dips shall be geometrically arranged so that the rigid body and significant flexible modes of the airplane can be selectively excited. The selected bumps or dips shall be taxied over so that the response will be due only to the known bump or dip. Single and multiple bumps or dips shall be traversed so as to cause both aircraft vertical translation and pitch-plunge motion. In addition, for airplanes designed for support area airfield operation, the bumps or dips shall be traversed perpendicular to the crest or trough line and at least one additional angle determined from the analysis.

4.2.3.2 Landing tests. The following landing tests shall be performed:

a. Two-point landing

b. Three-point landing

c. Tail-down landing

4.2.3.3 Arrestments. Arrestments shall be performed as follows:

a. Emergency abort of takeoff

b. Landing arrestment not including fly-in arrestments unless design criteria requirement.

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4.2.3.4 Braking. Braking shall be tested as follows:

a. Abrupt braking using maximum available force and force application rate at the brake pedals or controls

b. Maximum unsymmetrical braking

c. Maximum reverse braking.

4.2.3.5 Turning. Various turning angles at various taxi speeds shall be conducted, as applicable, to insure testing the critical condition.

4.2.3.6 Pivoting. The most critical design pivoting condition shall be tested.

4.2.3.7 Engine runup. The following engine runup tests shall be conducted:

a. Engines shall be run up to maximum symmetrical power and thrust using maintenance tiedown procedures

b. Engines shall be run up to maximum unsymmetrical power and thrust using maintenance tiedown procedures

c. Preflight. Engines shall be run up to maximum operational check power and thrust using operational holdback procedures.

4.2.3.8 Ski loads. Design ski loads conditions shall be tested.

4.2.3.9 Towing. Maximum towing conditions shall be tested.

4.3.2.10 Test required to define suspected new critical conditions. If analysis of the ground load data indicates the possibility of new critical conditions not previously investigated, additional tests shall be conducted.

4.3 Flight and ground operations demonstration. The test conditions for this demonstration shall include all the critical analytical loading conditions such that the aircraft will be insured of experiencing the maximum load attainable within its operating limitations. The test maneuvers specified under 4.5 shall be performed as applicable. This demonstration shall be conducted in the following two phases:

a. Initial phase tests to 80 percent limitations. These tests are essentially buildup tests prior to conducting the final phase tests. After completion of these tests, the recorded data and aircraft structure

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shall be inspected for any indication of a structural problem before proceeding to the final phase tests.

b. Final phase tests to 100 percent limitations. The final phase tests shall be performed after completion of the initial phase tests and with the specific approval of the procuring activity.

4.4 Dynamic response tests. The dynamic response tests shall include various flight and ground handling test conditions which simulate planned operational conditions such that the measured loads data may be used to correlate with and prove the aircraft dynamic loads analysis and analytical dynamic transfer functions. When possible, the dynamic response testing shall be combined with the flight and ground operations loads survey testing. The following test conditions shall be included, as applicable, by the contractor in preparing the test program:

4.4.1 Test flights. Flights shall be conducted through turbulence having a minimum rms true gust velocity of 2 feet per second or having a level sufficient to provide data for correlation with the dynamic analysis, whichever is greater. The speed, altitude, weight, and configuration combinations shall be those where significant response is indicated by the dynamic analysis used for the calculations of the fatigue and design gust loads.

4.4.1.1 Flight attitude. The test airplane shall be flown straight and level over the test course with minimum control inputs.

4.4.1.2 Test period time. To obtain adequate data for substantiating the analysis, 3 minutes of flight test data shall be required for each test condition. However, if this data is not sufficient to substantiate the analysis, additional data shall be measured to do so.

4.4.2 Landing tests. Landing tests shall be conducted at various landing speeds, sinking speeds, and gross weight combinations where symmetrical response is indicated by the dynamic analysis used for the calculation of the fatigue and design landing loads. However, at least one landing shall be conducted at the critical condition/configuration and at no less than 80 percent of the design landing sink rate.

4.4.3 Taxi tests. Taxi tests shall include, as applicable, taxiing on operational surfaces having degrees of roughness as specified in the applicable contractual document. The surfaces shall be traversed in both directions for the specified taxi distances at the required velocities, weights, and other applicable parameters needed to provide sufficient data to validate the airframe dynamic loads analysis. The data shall be reduced using power spectral techniques or techniques specified by the procuring

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activity. The power spectral density input shall be obtained from a survey of the runways using an Air Force profilograph, or equivalent.

4.4.4 In-flight refueling tests. For those airplanes capable of in-flight refueling, tests shall be conducted with the airplane loadings, configurations, altitudes, and speeds that are representative of service operations and speeds that are representative of service operations and contract specifications. Airplane structural response and airplane refueling equipment and local structure loads and stresses shall be measured.

4.5 Test maneuver requirements. Test maneuver requirements for the various tests are specified below.

4.5.1 Normal symmetrical pullup. At the required test altitude and required test speed, a normal (see 6.2.2) symmetrical pullup shall be performed to obtain the required test load factor or, if less, the maximum load factor within the maximum tolerable buffeting (4.6.4).

4.5.2 Normal symmetrical pushdown. At the required test altitude and required test speed, a normal symmetrical pushdown shall be performed to obtain the required test load factor or, if less, the maximum load factor within the maximum tolerable buffeting (4.6.4). When this maneuver is required at flight altitudes below 10,000 feet, a pullup to a sufficient climb angle shall be performed first in order to avoid excessive dive angles after carrying out the pushdown.

4.5.3 Normal uncoordinated rolling pullout. At the required test altitude and required test speed, a normal uncoordinated rolling pullout shall be performed. After establishing a steady right turn at a bank angle which corresponds to the required test load factor, the pullout shall be to the left. Where total load instrumentation is not provided on the left wing, the pullout shall be performed in both directions. The airplane shall be rolled through an angle equal to two times the initial bank angle. Aileron control forces shall be applied until the rate of roll is built up to a maximum. The roll shall be checked by the application of oppositely directed control forces so that the specified bank angle will not be exceeded. During the initial phase tests, the rate of application of aileron control forces shall be the maximum but shall not exceed that which placed 80 percent of the limit load on the major structural components of the airplane. The rudder shall be held fixed in its trim position as established by the maneuver entry speed. The elevator or stabilator position shall be held constant, unless change is required to avoid exceeding the required test load factor.

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4.5.4 Normal uncoordinated 360-degree roll (fighter and trainer aircraft only). The initial load factor shall be 1.0. At the required test altitude and required test speed, a normal uncoordinated 360-degree roll shall be performed to the left. Where total load instrumentation is not provided on the left wing, the roll shall be performed in both directions. Starting from a wings-level flight attitude and a load factor of 1.0, the airplane shall be put into a 360-degree roll. Aileron control forces shall be applied until the rate of roll is built up to a maximum (4.6.3). The roll shall be checked by the application of oppositely directed control forces to avoid exceeding the specified angle of roll. During initial phase tests, the rate of application aileron control forces shall be the maximum but shall not exceed that which places 80 percent of limit load on the major structural components of the airplane. The rudder shall be held fixed in its trim position as established by the maneuver entry speed. The elevator or stabilator position shall be held constant, unless change is required to avoid exceeding the required test load factor.

4.5.5 Steady-state yaw maneuver. At the required test altitude and required test speed, a level yaw maneuver to maximum steady-state sideslip angle shall be performed. A rudder pedal force of 300 pounds, full rudder deflection, or maximum output of the power control system shall be used while maintaining the airplane at a wings-level flight attitude throughout the maneuver.

4.5.6 Abrupt symmetrical pullup. At the required test altitude and required test speed, an abrupt (see 6.2.3) symmetrical pullup shall be performed. To avoid exceeding the maximum tolerable buffeting (4.6.4), the pilot may check this maneuver as soon as the movement applied to the control surfaces is equal to that required for the specified test load factor. Abrupt application of control forces shall be in accordance with 4.6.5.

4.5.7 Abrupt symmetrical pullup with abrupt checking. At the required test altitude and required test speed, an abrupt symmetrical pullup with abrupt checking shall be performed to obtain the required test load factor. For this maneuver, the rate of elevator motion shall be the maximum obtainable; without exceeding the test load factor. The maneuvers shall be performed in accordance with MIL-A-8861. Abrupt application of control forces shall be in accordance with 4.6.5.

4.5.8 Abrupt symmetrical pushdown with abrupt checking. At the required test altitude and required test speed, an abrupt symmetrical pushdown with abrupt checking shall be performed to obtain the required test load factor. For this maneuver, the rate of elevator motion shall be the maximum obtainable, without exceeding the test load factor. The maneuvers shall be performed in accordance with MIL-A-8861. Abrupt application of control forces shall be in accordance with 4.6.5.

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4.5.9 Abrupt coordinated rolling pullout. At the required test speed, an abrupt coordinated rolling pullout shall be performed. After establishing a steady right turn at a bank angle which corresponds to the required test load factor, the pullout shall be to the left. The initial load factor shall be a range of values between 1.0 and $0.8n_z$. The airplane shall be rolled through an angle equal to two times the initial bank angle. Aileron control forces shall be abruptly applied until the rate of roll is built up to a maximum (4.6.3). The roll shall be abruptly checked with oppositely directed control forces so that the specified bank angle will not be exceeded. During the roll initiation, the elevator or stabilator position shall be held constant and aileron and rudder applied as necessary to coordinate the maneuver. Abrupt application of control forces shall be in accordance with 4.6.5.

4.5.10 Abrupt uncoordinated rolling pullout with abrupt checking. At the required test altitude and required test speed, an abrupt uncoordinated rolling pullout shall be performed. After establishing a steady right turn at a bank angle which corresponds to the required test load factor, the pullout shall be to the left. The initial load factor shall be a range of values between 1.0 and $0.8n_z$. The airplane shall be rolled through an angle equal to two times the initial bank angle. Aileron control forces shall be abruptly applied until the rate of roll is built up to a maximum (4.6.3). The roll shall be abruptly checked with oppositely directed control forces so that the specified bank angle will not be exceeded. During the roll phase of this maneuver, the rudder shall be held fixed in its trim position as established by the maneuver entry speed. The elevator or stabilator position shall be held constant unless change is required to avoid exceeding the required test load factor. Abrupt application of control forces shall be in accordance with 4.6.5.

4.5.11 Abrupt coordinated 180-degree roll (fighter and trainer aircraft only). At the required test altitude and required test speed, an abrupt coordinated 180-degree roll shall be performed with fighter and trainer airplanes. The load factor shall be a range of values between +1.0 and -1.0 with the maneuver starting from a wings-level flight altitude and then executing a 180-degree roll. Aileron control forces shall be abruptly applied until the rate of roll is built up to a maximum (4.6.3). During the roll phase of this maneuver, the elevator or stabilizer position to entry shall be held constant and only the rudder changed as necessary to coordinate the maneuver. Abrupt application of control forces shall be in accordance with 4.6.5.

4.5.12 Abrupt uncoordinated 180-degree roll (fighter and trainer aircraft only). At the required test altitude and required test speed, an abrupt uncoordinated 180-degree roll shall be performed with fighter and trainer

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airplanes. Starting with a wings-level flight attitude and a load factor of 1.0, the airplane shall be rolled to the left by an abrupt application of the aileron control force and then checking the roll with an abrupt application of an oppositely directed control force to avoid exceeding the specified 180-degree bank angle. Aileron control forces shall be abruptly applied until the rate of roll is built up to a maximum (4.6.3). During the roll, the rudder shall be held fixed in its trim position as established by the maneuver entry speed. The elevator or stabilator position shall be held constant unless change is required to avoid exceeding the required test load factor. Abrupt application of control forces shall be in accordance with 4.6.5.

4.5.13 Abrupt uncoordinated 360-degree roll (fighter and trainer aircraft only). The initial load factor shall be 1.0. At the required test altitude and required test speed, an abrupt uncoordinated 360-degree roll shall be performed to the left. Starting from a wings-level flight attitude and an initial load factor of 1.0, the airplane shall be put into a 360-degree roll. Aileron control forces shall be abruptly applied until the rate of roll is built up to a maximum (4.6.3). The roll shall be checked by the abrupt application of oppositely directed control forces to avoid exceeding the specified angle of roll. During the roll, the rudder shall be held fixed in its trim position as established by the maneuver entry speed. The elevator or stabilator position shall be held constant unless change is required to avoid exceeding the unsymmetrical load factor limits. Abrupt application of control forces shall be in accordance with 4.6.5.

4.5.14 Landing approach pullup. At the required test altitude and required test speed, a symmetrical pullup shall be performed to obtain the required test load factor or, if less, the maximum obtainable load factor in the full-flaps, gear-down landing approach configuration.

4.5.15 Landing approach rolling pullout. At the required test altitude and required test speed, an abrupt coordinated roll shall be performed. Starting from a wings-level flight attitude and a load factor of 1.0, the roll shall be to the left by abruptly applying oppositely directed control forces. For cargo and bomber class II airplanes, the bank angle shall not exceed 60 degrees and for all other types of airplanes, the bank angle shall not exceed 90 degrees. Rudder and elevator or stabilator forces shall be applied as required to coordinate the maneuver. Abrupt application of control forces shall be in accordance with 4.6.5.

4.5.16 Rudder maneuver for high-speed steady sideslip with rudder reversed (fighter and trainer aircraft only). At the required test altitude and required test speed, a steady sideslip maneuver with an abrupt reversal

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of the rudder position shall be performed with fighter and trainer airplanes. A pilot effort of 180 pounds or full rudder deflection, whichever occurs first, shall be initially applied to the left rudder control. When the airplane reaches the maximum steady sideslip angle, the rudder shall be reversed using a pilot effort of 180 pounds or full rudder deflection, whichever occurs first. Full reversal of the rudder completes this maneuver and to avoid excessive loads on the airplane, the rudder should be returned to neutral position for immediate recovery. During this maneuver, the aileron or spoiler controls shall be used to keep the airplane in a wings-level flight attitude. Abrupt application of control forces shall be in accordance with 4.6.5.

4.5.17 Rudder kick maneuver with abrupt return. At the required test altitude and required test speed, an abrupt rudder kick to dynamic overswing with an abrupt return from steady sideslip attitude shall be performed. A pilot effort of 180 pounds or full rudder deflection, whichever occurs first, shall be initially applied to the left rudder control. The maximum rudder angle shall be held until the airplane has reached dynamic overswing and returned to a steady sideslip attitude. Once this is accomplished, the rudder shall be abruptly returned to the neutral position. During this maneuver, the aileron or spoiler controls shall be used to keep the airplane in a wings-level flight attitude. Abrupt application of control forces shall be in accordance with 4.6.5.

4.5.18 Rudder kick maneuver for landing approach. At the required test altitude and required test speed, an abrupt rudder kick to dynamic overswing with an abrupt return from steady sideslip attitude shall be performed with the airplane in a flaps and gear full-down landing approach configuration. A pilot effort of 300 pounds or full rudder deflection, whichever occurs first, shall be initially applied to the left rudder control. The maximum rudder angle shall be held until the airplane has reached dynamic overswing and returned to a steady sideslip attitude. Once this is accomplished, the rudder shall be abruptly returned to the neutral position. During this maneuver, the aileron or spoiler controls shall be used to keep the airplane in a wings-level flight attitude. Abrupt application of control forces shall be in accordance with 4.6.5.

4.5.19 Unsymmetrical power simulation maneuver. Depending upon the most critical power failure condition for vertical tail loads, as indicated by analytical calculations, an unsymmetrical power simulation maneuver shall be performed. At the required test altitude and required test speed, an unsymmetrical power simulation maneuver shall be performed by a very rapid reduction in power on the critical left engine while all other engines are delivering their normal rated power for the test condition. The reduction to minimum power shall be as rapid as possible. The rudder, initially in its trim position, shall be moved by a 300-lb force applied in 0.2

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second or at the maximum rate of the system in order to restore the airplane to a zero yaw condition. The restorative force shall be applied at the time from instant of failure determined to be critical from analytical calculation. Meanwhile, the aileron or spoiler controls shall be used to keep the airplane in a wings-level flight attitude.

4.5.20 Deceleration device extension maneuver. Speed brakes, dive flaps, maneuvering flaps, and any other deceleration devices shall be extended at limit speed (V_L).

4.6 Miscellaneous requirements. Miscellaneous requirements are as follows:

4.6.1 Loading. Loading includes the design loads of expendable articles and fixed equipment except that replaced by instrumentation or test equipment. Where design loads include items not available at the time of the test, ballast shall be used to simulate these design loads. If the actual weight is greater due to instrumentation requirements than that specified in the design and contract specifications, the actual weight shall be used and the test load factors shall be adjusted to make allowance for the overweight condition. Airplanes shall be loaded so that maneuvers may be performed with the critical loading at the test altitude. At the time maneuvers are performed, a variation in the fuel loading, equivalent to ±5 percent of the required airplane weight for test, is allowed but such variation shall require corresponding adjustment of the test load factors. Actual airplane weight and balance data shall be determined and submitted as a part of the test data for each maneuver. This loading requirement shall pertain to Final Phase Tests. For the Initial Phase tests, any test weight mutually agreed upon between the contractor and procuring activity may be used and the results corrected to the required airplane loading.

4.6.1.1 Fuel loading. Fuel loading shall be as follows:

a. If a maximum of internal fuel is required, the fuel loading shall be the full internal fuel capacity less the minimum amount of fuel required from engine start or in-flight refueling to the test altitude. This is applicable to clean configuration requirements and to full external tank requirements.

b. If full internal fuel is required, the fuel loading shall be the full internal fuel capacity plus the amount of fuel required from engine start or in-flight refueling to the test altitude. If external tanks are carried, the overage shall be carried in the external tanks. Before the test maneuvers begin, these external tanks shall be empty.

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c. If a minimum of internal fuel is required, the fuel loading shall be as specified in MIL-A-8860 but in no case less than the fuel required to sustain 30 minutes of operation at cruising power.

d. If full external tanks are required, the external tanks shall be maintained full throughout the test maneuvers on airplanes with selective fuel controls. On airplanes with automatic or no selective external fuel controls, the external tank fuel loading shall be the full tank capacity less the minimum amount of fuel required from engine start or in-flight fueling to the test altitude.

4.6.2 Trim requirements. For maneuvers, the airplane shall be trimmed for zero forces on all controls at the following times:

a. On symmetrical pullups, just prior to starting pullup.

b. On rolling pullouts, before the maneuver is initiated.

c. On pushdowns, just prior to starting pushdown.

d. Rudder maneuvers, just prior to application of rudder force. Once these trim conditions are established, they shall not be changed during the maneuver. However, where these trim conditions prove impractical, other trim conditions may be used to accomplish the required maneuver and shall be clearly defined in the flight test report.

4.6.3 Rate of roll requirements. For rolling maneuvers, the maximum rate of roll shall be that obtained within the specified angle of bank by applying either a 60-pound lateral control force (two equal and opposite 48-pound forces applied at the circumference of the control wheel), full stick or wheel displacement, or full aileron or spoiler deflection, whichever occurs first. The roll rate shall be the maximum design roll rate of the airplane. The roll shall be checked with an oppositely directed force so that the specified angle of bank will not be exceeded. The rate of application of the aileron force shall be as specified in 4.6.5.

4.6.4 Buffeting requirements. Maximum tolerable buffeting (see 6.2.4) shall be verified by the procuring activity after a study of the test data has been made.

4.6.5 Abrupt application of control forces. Abrupt application of control forces shall be in accordance with the structural design criteria or as follows:

a. For pitching maneuvers:
   0.2 second on A, F, TF, T, and O types

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0.3 second on U, B_1, B_{11}, and C assault types

0.4 second on all other types

b. For rolling maneuvers:

0.1 second on stick-type controls

0.3 second on wheel-type controls

c. For yawing maneuvers:

0.3 second on C and B_{11} types

0.2 second on all other types

5. PREPARATION FOR DELIVERY

This section is not applicable to this specification.

6. NOTES

6.1 Intended use. This specification is intended to be used for structural flight and ground operation testing.

6.2 Definitions. For the purpose of this specification, the following definitions shall apply:

6.2.1 Test speed. A test speed is defined as the speed attained for the required load factor and sideslip angle.

6.2.2 Normal maneuver. A normal maneuver is defined as a relatively smooth maneuver in which acquisition of basic data is a primary requirement.

6.2.3 Abrupt maneuver. An abrupt maneuver is defined as a maneuver in which the abrupt application of control forces is a primary requirement.

6.2.4 Maximum tolerable buffeting. Maximum tolerable buffeting is defined as that buffeting which produces an incremental load factor (half amplitude) at the center-of-gravity of the airplane equivalent to 5 percent of the required test load factor.

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6.3 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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